

Overblik over Hesel ligningerne

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March 2026

1 Standard form

HESEL ligningerne er givet på følgende form:

$$\begin{aligned} d_t n + n \mathcal{K}(\phi) - \mathcal{K}(p_e) &= \Lambda_n \\ d_t^0 \omega^* + \{\phi, p_i\} - \mathcal{K}(p_e + p_i) &= \Lambda_{\omega^*} \\ \frac{3}{2} d_t p_e + \frac{5}{2} p_e \mathcal{K}(\phi) - \frac{5}{2} \mathcal{K}\left(\frac{p_e^2}{n}\right) &= \Lambda_{p_e} \\ \frac{3}{2} d_t p_i + \frac{5}{2} p_i \mathcal{K}(\phi) + \frac{5}{2} \mathcal{K}\left(\frac{p_i^2}{n}\right) - p_i \mathcal{K}(p_e + p_i) &= \Lambda_{p_i} \end{aligned}$$

2 Lambda termer

Lambda termerne er givet ved:

$$\begin{aligned} \Lambda_n &= D_e \nabla_{\perp}^2 n - \frac{n}{\tau} - \frac{n - n_p}{\tau_p} - \alpha \left(\tilde{T}_e + \frac{\tilde{T}_e}{\tilde{n}} \tilde{n} - \tilde{\phi} \right) \\ \Lambda_{\omega^*} &= D_i \nabla_{\perp}^2 \omega^* - \frac{\omega^*}{\tau} + \frac{\rho_s}{L_c} \left[1 - \exp \left(\phi_m - \frac{\phi_s}{T_{e,s}} \right) \right] \\ &\quad - \alpha \left(\tilde{T}_e + \frac{\tilde{T}_e}{\tilde{n}} \tilde{n} - \tilde{\phi} \right) \\ \Lambda_{p_e} &= \frac{5}{2} D_e \nabla_{\perp}^2 p_e + \left(\frac{16}{6} - \frac{5}{2} \right) \nabla \cdot (n \nabla_{\perp} T_e) - \frac{9p_e}{2\tau} - \frac{T_e}{\tau^{SH}} - \Theta - \frac{p_e - p_{e,p}}{\tau_p} \\ &\quad - \bar{T}_e \alpha \left(\tilde{T}_e + \frac{\tilde{T}_e}{\tilde{n}} \tilde{n} - \tilde{\phi} \right) \\ \Lambda_{p_i} &= D_i \nabla_{\perp}^2 p_i - D_i T_i \nabla_{\perp}^2 n - \frac{9p_i}{2\tau} + \Theta - \frac{p_i - p_{i,p}}{\tau_p} + p_i \Lambda_{\omega^*} \end{aligned}$$

3 Funktioner

Funktionerne er givet ved:

$n : n(x, y, t) :$	Tæthed
$p_{e,i} : p_{e,i}(x, y, t) :$	Elektron- og iontryk
$T_{e,i} : T_{e,i}(x, y, t) = p_{e,i}/n :$	Elektron- og iontemperatur
$\phi : \phi(x, y, t) :$	Elektrisk potentiale
$\omega^* : \omega^*(x, y, t) = \nabla^2 \phi + \nabla^2 p_i :$	Vorticitet

4 Operatorer

Herved er operatorerne defineret med:

$$\mathrm{d}_t^0 f = \partial_t f + \{\phi, f\}$$

$$\mathrm{d}_t f = \partial_t f + \frac{1}{B} \{\phi, f\}$$

$$\{f, g\} = \partial_x f \partial_y g - \partial_y f \partial_x g$$

$$\mathcal{K}(f) = -\frac{\rho_s}{R} \partial_y f$$

$$\bar{f} = \frac{1}{L_y} \int_0^{L_y} f \mathrm{d}y$$

$$\tilde{f} = f - \bar{f}$$

$$\nabla = (\partial_x, \partial_y, \partial_z)^T$$

$$\nabla_\perp = (\partial_x, \partial_y, 0)^T$$

$$\nabla^2 = (\partial_x^2, \partial_y^2, \partial_z^2)^T$$

$$\nabla_\perp^2 = (\partial_x^2, \partial_y^2, 0)^T$$

5 Koeffecienter

Samling af koeffecienter:

$$\begin{aligned}
\rho_s &= \sqrt{\frac{T_{e0}}{m_i \Omega_{ei}^2}} \\
D_i &= \left(1 + \frac{R}{a} q^2\right) \frac{\rho_{i0}^2 v_{ii,0}}{\rho_{s0}^2 \omega_{i0}} \\
D_e &= \left(1 + \frac{T_{i0}}{T_{e0}}\right) \left(1 + \frac{R}{a} q^2\right) \frac{\rho_{e0}^2 v_{ei,0}}{\rho_{s0}^2 \omega_{i0}} \\
\Theta &= \frac{3m_e}{m_i} v_{ei} (p_e - p_i) \\
\tau &= \frac{L_b}{2M_{\parallel} c_s} \\
\tau^{SH} &= \frac{L_b}{X_{\parallel,e}^{SH}} \\
\alpha &= \frac{2T_{e0}}{v_{ei} m_e L_{\parallel}^2 \omega_{i0}} \\
c_s &= \sqrt{(T_e + T_i)/m_i} \\
\phi_m &= \ln(\sqrt{m_i/2\pi m_e}) \\
L_{\parallel} &= L_b = qR
\end{aligned}$$

6 parametre

Parametre for HESEL ligningerne:

$B_0 = 1.9T$	Magnetfeltstyrke
$R = 1.65m$	Tokamakens store radius
$a = 0.5m$	Tokamakens lille radius
$q = 5.32$	Sikkerhedsfaktor
$L_b = qR \approx 8m$	Poloidal angle
$L_c \approx 15m$	Divertor leg along the magneticfield lines
$L_c = 2.5m$	Divertor leg shadow region
$T_i = T_e = 25eV$	Used for calculating cs
$M_{\parallel} = 0.5$	Mach tal
$2M_{\parallel} c_s \approx 50000m/s$	plasma radially propagation speed
$X_{\parallel,e}^{SH} = 3.16V_e^2/v_{ee}$	Spitzer-Härm heat diffusivity

Afhængige parametre:

Referenceværdier (lav kollisionsalitet, ved LCFS):

$$n_0 = 1.5 \cdot 10^{19} \text{ m}^{-3}$$

$$T_{e0} = 25 \text{ eV}$$

$$T_{i0} = 25 \text{ eV}$$

$$\Lambda_0 = 0.758$$

Referenceværdier (høj kollisionsalitet):

$$n_0 = 3.5 \cdot 10^{19} \text{ m}^{-3}$$

$$T_{e0} = 15 \text{ eV}$$

$$T_{i0} = 15 \text{ eV}$$

$$\Lambda_0 = 4.6$$

Numerisk domæne:

$$L_x = L_y = 95.0 \text{ mm}$$

$$L_{\text{edge}} = 34.4 \text{ mm}$$

$$L_{\text{SOL}} = 47.5 \text{ mm}$$

$$L_{\text{wall}} = 12.3 \text{ mm}$$

$$N_x = N_y = 1024$$